COMMERCIAL ITEM DESCRIPTION

Radiacmeter, Pocket-Sized, Digital Alarm, Dosimeter

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This document covers the basic requirements for a pocket-sized Digital Alarm Dosimeter (DAD) for use by Air Force Non-destructive Inspection (NDI) personnel. The dosimeter is required to operate under extreme temperature and humidity conditions. The instrument could be physically subjected to accidental drops from the wing of an aircraft, contact with caustic chemicals, and other hazards commonly encountered in the Air Force maintenance environment. The dosimeter shall be extremely rugged and light-weight. The dosimeter shall be capable of continuously monitoring the rate of exposure and recording the accumulated dosage. The dosimeter shall be capable of measuring X-radiation in the range of 40 Kev to 1.2 Mev. The dosimeter must accurately measure the received radiation, store the accumulated dose, and provide a digital readout on a light emitting diode (LED) display. The dosimeter must have an alarm signal which will produce an audible signal when a given dose has been accumulated and emit an audible signal that is adjustable or in direct proportion to the radiation intensity the person wearing the dosimeter is being exposed.

Salient characteristics. The dosimeter shall have at least the following performances and components:

1. Environmental. The dosimeter shall be operatable at temperatures of -20° C to +50° C (-4° F to +122° F) and at 0 to 95% relative humidity.

2. Ruggedness. The dosimeter shall be capable of withstanding a 2 meter(m)(78 inch (in)) drop test onto a flat concrete surface three consecutive times and exhibit satisfactory performance. The dosimeter will be oriented in mutually perpendicular planes to concrete surface.

3. Response characteristics. The dosimeter shall meet the following requirements:

a. Radiation detected. The dosimeter shall be capable of detecting Gamma and X-ray radiation at the range of 40 Kev to 1.2 Mev.

b. Range. The dosimeter shall display the accumulated dosage in milliroentgen (mR). Minimum range shall be 0 - 999 mR.

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c. Detector type. The dosimeter shall be provided with Halogen-Quenched, energy-compensated GM tube. Internal background from GM tube shall be less than 2 mR/day.

d. Energy dependence. Energy dependence of the dosimeter shall be linear within ±30% in the range of 40 Kev to 1.2 Mev.

e. Accuracy. The dosimeter shall have an accuracy of ±15% over a range of 0.1 to 1 R/hr referenced to Cs-137.

f. Audible signal (chirp). The audible signal shall have 1 chirp per 0.25 mR minimum and have an adjustable chirp rate or emit chirps in direct proportion to the radiation intensity.

g. Alarm point. The alarm points should be either variable or fixed. If a variable alarm point is used, it should be adjustable over a range of 4-32mR as a minimum (may be greater) with several increments within selected range. Adjustment must be external and maybe screwdriver adjustable. If fixed alarm point is used, recommend no less than 10 mR and no more than 16 mR. Exposure rate alarm shall continue to operate in fields up to 1000 times the highest alarm set point.

h. Alarm signal. The alarm signal shall have a sound pressure level of 75 dBA in free air at a distance of 30 centimeter (cm) (11.8 in.) from the dosimeter.

i. Exposure accumulation. The exposure accumulation shall continue when the alarm is activated.

4. Controls. Controls on the dosimeter shall be designed to prevent inadvertent changing of control setting. No control or combination of controls will be able to reset the total stored (memory) exposure reading except the one button whose design is to accomplish reset.

5. Memory reset. The total stored exposure (memory) shall require the use of an auxiliary tool (toothpick for example) to accomplish reset

6. Display. The dosimeter shall have a non-continuous LED display with a manual readout activator that will not reset the total exposed memory. Display shall remain on at least five seconds after activation or as long as manual readout button is pushed. The display shall have a 3 or 4 digit readout for displaying the entire range of total exposure in mR.

7. Dimensions. The dimensions of the dosimeter shall not exceed the following requirements:

Length: 12.7 cm (5 in)
Width: 6.67 cm (2.63 in)
Thickness: 3.3 cm (1.3 in)

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8. Weight. The weight of the dosimeter shall not exceed 227 grams (8 ounces) with battery(ies).

9. Clip. The dosimeter shall be provided with a clip that is capable of fastening to cloth pocket and/or belt. The gripping power shall be sufficient to hold the dosimeter in place when a constant force equal to three times the weight of the dosimeter is applied.

10. Case. The dosimeter case shall be made of an impact and corrosion resistant material (example: ABS molded plastic). If a light weight metal is used, it shall have a corrosion-resistant surface coating.

11. Batteries. The dosimeter shall operate on a 9 volt alkaline or equivalent battery. The battery life expectancy shall be greater than 100 hours when used continuously in a 10 mR/hr field.

12. Low battery indicator. The low battery indicator shall be able to indicate a reserve of at least 8 hrs in a 100 mR/hr field.

13. Memory. The total exposure memory information shall be maintained for an additional 2 hrs beyond the 8 hrs required as a reserved after low battery is indicated. Backup battery design may be incorporated to accomplish this requirement. Between uses, the dosimeter shall be capable of being turned off without losing the stored dose. The memory will erase only by depressing a reset button or switch that is protected against accidental resetting.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. Businesses that can not offer products with established commercial market acceptability, may compete by submission of bid samples.

Regulatory requirements. The vendor/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Preservation, packaging, packing, labeling and marking. The preservation, packaging, packing, labeling and marking of the dosimeter shall be as specified in the contract or order.

Custodian activities:
Air Force - 99
Army - EA
Navy - SH

Preparing activity:
Air Force - 82

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